



## **INDIA RANKINGS 2024**

### **NATIONAL INSTITUTIONAL RANKING FRAMEWORK**

**Methodology for Ranking of Academic Institutions in India**

**(RANKING METRICS FOR ENGINEERING)**



**Ministry of Human Resource Development**

## Summary of Ranking Parameters and Weightages- 2024

(Engineering)

| Sr. No. | Parameter                          | Marks | Weightage |
|---------|------------------------------------|-------|-----------|
| 1       | Teaching, Learning & Resources     | 100   | 0.30      |
| 2       | Research and Professional Practice | 100   | 0.30      |
| 3       | Graduation Outcomes                | 100   | 0.20      |
| 4       | Outreach and Inclusivity           | 100   | 0.10      |
| 5       | Perception                         | 100   | 0.10      |

### Engineering

| S.No. | Parameters                                                                                                                                                                                                                                                                                             | Marks      |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| 1.    | <b>Teaching, Learning &amp; Resources (TLR)</b><br><b>Ranking weight: 0.30</b>                                                                                                                                                                                                                         | <b>100</b> |
|       | A. Student Strength including Doctoral Students(SS): 20 marks<br>B. Faculty-student ratio with emphasis on permanent faculty (FSR): 30 marks<br>C. Combined metric for Faculty with PhD (or equivalent) and Experience (FQE): 20 marks<br>D. Financial Resources and their Utilisation (FRU): 30 marks |            |
| 2.    | <b>Research and Professional Practice (RP)</b><br><b>Ranking weight: 0.30</b>                                                                                                                                                                                                                          | <b>100</b> |
|       | A. Combined metric for Publications (PU): 35 marks<br>B. Combined metric for Quality of Publications (QP): 40 marks<br>C. IPR and Patents: Published and Granted (IPR): 15 marks<br>D. Footprint of Projects and Professional Practice (FPPP): 10 marks                                                |            |
| 3.    | <b>Graduation Outcomes (GO)</b><br><b>Ranking weight: 0.20</b>                                                                                                                                                                                                                                         | <b>100</b> |
|       | A. Combined metric for Placement and Higher Studies (GPH): 40 marks<br>B. Metric for University Examinations (GUE): 15 marks<br>C. Median Salary (GMS): 25 marks<br>D. Metric for Number of Ph.D. Students Graduated (GPHD): 20 marks                                                                  |            |
| 4.    | <b>Outreach and Inclusivity (OI)</b><br><b>Ranking weight: 0.10</b>                                                                                                                                                                                                                                    | <b>100</b> |
|       | A. Percentage of Students from other States/Countries (Region Diversity RD): 30 marks<br>B. Percentage of Women (Women Diversity WD): 30 marks<br>C. Economically and Socially Challenged Students (ESCS): 20 marks<br>D. Facilities for Physically Challenged Students (PCS): 20 marks                |            |
| 5.    | <b>Perception (PR)</b><br><b>Ranking weight: 0.10</b>                                                                                                                                                                                                                                                  | <b>100</b> |
|       | A. Peer Perception: Employers & Academic Peer (PR): 100 marks                                                                                                                                                                                                                                          |            |

## **1. Teaching, Learning & Resources (TLR): 100 marks**

- **Ranking weight: 0.30**

- **Overall Assessment Metric:**

$$\text{TLR} = \text{SS (20)} + \text{FSR (30)} + \text{FQE (20)} + \text{FRU (30)}$$

- **Component metrics based on :**

**A. Student Strength including Ph.D. Students: SS**

**B. Faculty-Student Ratio with emphasis on permanent faculty: FSR**

**C. Combined metric for Faculty with PhD (or equivalent) and Experience:  
FQE**

**D. Financial Resources and Their Utilisation: FRU**

**A. Student Strength including Ph.D. students (SS): 20 Marks**

- $SS = f(N_T, N_E) \times 15 + f(N_P) \times 5$
- The functions  $f(N_T, N_E)$  and  $f(N_P)$  are functions to be determined by NIRF.
- $N_T$ : Total sanctioned approved intake in the institution considering all UG and PG programs of the institution.
- $N_E$ : Total number of students enrolled in the institution considering all UG and PG Programs of the institution.
- $N_P$  = Total number of students enrolled for the doctoral program till previous academic year.
- **Primary Data: To be provided in a prescribed Format.**

**B. Faculty-Student Ratio with emphasis on permanent faculty (FSR): 30 marks**

- **$FSR = 30 \times [15 \times (F/N)]$**
- $N = N_T + N_p$
- F: Full time regular faculty in the institution in the previous year.
- Regular appointment means Faculty on Full time basis. *Faculty on contract basis/ad-hoc basis will be considered if the concerned faculty has taught in both the semesters of academic year 2022-23.*
- Faculty members with Ph.D. and M.E./M.Tech. will be considered and counted here. Faculty member with a Bachelor's degree will not be counted.
- Expected ratio is 1:15 to score maximum marks.
- **For  $F/N < 1$ : 50, FSR will be set to zero.**
- ***Primary Data: Faculty List to be provided in the Prescribed Format.***

**C. Combined Metric for Faculty with PhD (or equivalent) and Experience (FQE): 20 marks**

- $FQ = 10 \times (F_{RA}/95)$  ,  $F_{RA} < 95\%$ ;
- $FQ = 10$ ,  $F_{RA} \geq 95\%$ .
- Here  $F_{RA}$  is the percentage of Faculty with Ph.D. (or equivalent qualification) with respect to the total no. of faculty required or actual faculty whichever is higher, in the previous year.

F1=Fraction with Experience up to 8 years;

F2= Fraction with Experience between 8+ to 15 years;

F3=Fraction with Experience > 15 years.

- $FE = 3\min(3F1, 1) + 3 \min(3F2, 1) + 4 \min(3F3, 1)$

Rationale: Full marks for a ratio of 1:1:1

- $FQE = FQ + FE$
- **Primary Data: Faculty List in the Prescribed Format.**

**D: Financial Resources and their Utilisation (FRU): 30 Marks**

- **FRU =  $7.5 \times f(\text{BC}) + 22.5 \times f(\text{BO})$**
- BC: Average Annual Capital Expenditure per student for previous three years pertaining to engineering discipline only. **(Excluding expenditure on construction of new buildings)**
- BO: Average Annual Operational(or Recurring) Expenditure per student for previous three years pertaining to engineering discipline only. **(Excluding maintenance of hostels and allied services)**
- *Primary Data: Figures in prescribed format for each.*

## **2.Research and Professional Practice (RP): 100 marks**

- **Ranking weight: 0.30**

- **Overall Assessment Metric:**

$$\text{RP} = \text{PU}(35) + \text{QP}(40) + \text{IPR}(15) + \text{FPPP}(10)$$

- **The component metrics explained on following pages.**

**A. Combined Metric for Publications: PU**

**B. Combined Metric for Quality of Publications: QP**

**C. IPR and Patents: Patents Published and Granted: IPR**

**D. Footprint of Projects and Professional Practice: FPPP**

**A. Combined metric for Publications (PU): 35 marks**

- **PU** =  $35 \times f(P/F_{RQ})$
- *P* is weighted number of publications as ascertained from suitable third party sources.
- *F<sub>RQ</sub>* is the maximum of nominal number of faculty members as calculated on the basis of a required FSR of 1:15 or the available faculty in the institution.
- **Sources: Third party sources.**

**B. Combined metric for Quality of Publications (QP): 40 Marks**

- $QP = 20 \times f(CC/F_{RQ}) + 20 \times f(TOP25P/P)$
- Here CC is Total Citation Count over previous three years.
- P is as computed for PU.
- TOP25P: Number of citations in top 25 percentile averaged over the previous three years.
- $F_{RQ}$  is the maximum of nominal number of faculty members as calculated on the basis of a required FSR of 1:15 or the available faculty in the institution.
- *Primary Data: Third Party Sources.*

**C. IPR and Patents: Patents Published and Granted (IPR): 15 marks**

- **$IPR = IPG + IPP$**

**$IPG = 10 \times f(PG)$**

- PG is the number of patents granted over the previous three years.

**$IPP = 5 \times f(PP)$**

- PP: No. of patents published over the previous three years.
- ***Primary Data: Third Party Sources.***

**D. Footprint of Projects and Professional Practice (FPPP): 10 marks**

- **FPPP = FPR + FPC**
- **FPR =  $7.5 \times f$  (RF)**
- RF is average annual research funding earnings (amount actually received in rupees) per faculty at institute level in previous three years.
- **FPC =  $2.5 \times f$  (CF)**
- CF is average annual consultancy amount (amount actually received in rupees) per faculty at institute level in previous three years.

### **3. Graduation Outcome (GO):100 marks**

- **Ranking weight: 0.20**
- **Overall Assessment Metric:**
- **$GO = GPH(40) + GUE(15) + GMS(25) + GPHD(20)$**
- **The component metrics are explained on the following pages:**
  - A. Combined metric for Placement and Higher Studies: GPH**
  - B. Metric for University Examinations: GUE**
  - C. Median Salary: GMS**
  - D. Metric for Number of Ph.D. Students Graduated: GPHD**

**A. Combined Metric for Placement and Higher Studies (GPH): 40 marks**

- **$GPH = 40 \times (N_p/100 + N_{hs}/100)$**
- $N_p$  = Percentage of graduating students (in UG/ PG programs) placed in the previous three years.
- $N_{hs}$  = Percentage of graduating students (in UG/ PG programs) who have been selected for higher studies in the previous three years.
- ***Primary Data: To be provided in prescribed format.***

**B. Metric for University Examinations (GUE): 15 Marks**

- $GUE = 15 \times \min [(N_g/80), 1]$
- $N_g$  is the percentage of Students (as a fraction of the approved intake), averaged over previous three years, passing the respective university examinations in stipulated time for the program in which enrolled.
- *Primary Data: To be provided in the prescribed format.*

**C. Median Salary(GMS): 25 Marks**

- $GMS = 25 \times f (MS)$
- MS = Median salary of graduates (in UG/PG program) in the previous three years from an institution.
- *Primary Data: To be provided in the prescribed format.*

**D. Metric for Number of Ph.D Students Graduated (GPHD): 20 Marks**

- **GPHD =  $20 \times f(N_{\text{phd}})$**
- $N_{\text{phd}}$  = Average number of Ph.D students graduated (awarded Ph.D) over the previous three years.
- ***Primary Data: Number of graduating Ph.D. Students as reflected in the approved Annual Report/Convocation Report to be provided in the prescribed format.***

#### **4. Outreach and Inclusivity (OI): 100 marks**

- **Ranking weight: 0.10**
- **Overall Assessment Metric:  $OI = RD(30) + WD(30) + ESCS(20) + PCS(20)$**
- **The component metrics are explained on following pages:**
  - A. Percentage of Students from other States/ Countries (Region Diversity): RD**
  - B. Percentage of Women (Women Diversity): WD**
  - C. Economically and Socially Challenged Students: ESCS**
  - D. Facilities for Physically Challenged Students: PCS**

**A. Percentage of Students from other States/ Countries (Region Diversity RD): 30 marks**

- **RD =  $25 \times \text{fraction of total students enrolled from other states} + 5 \times \text{fraction of students enrolled from other countries}$**
- ***Primary Data: To be provided in the prescribed format.***

**B. Percentage of Women (Women Diversity WD): 30 marks**

- **$WD = 15 \times (N_{WS}/50) + 15 \times (N_{WF}/20)$**
- $N_{WS}$  are the percentage of Women students.
- $N_{WF}$  are the percentage of Women Faculty including the women members in senior administrative positions, such as Heads of Departments, Deans or Institute Heads.
- Expectation: 50% women students and 20% women faculty.
- ***Primary Data: To be provided in the prescribed format.***

**C.Economically and Socially Challenged Students (ESCS) : 20 marks**

- **ESCS =  $20 \times f(N_{esc})$**
- $N_{esc}$  is the percentage of UG students being provided full tuition fee reimbursement by the institution to pursue their degree programs.
- ***Primary Data: To be provided by the institution in the prescribed format.***

**D. Facilities for Physically Challenged Students (PCS): 20 marks**

- **PCS = 20 marks, if the Institute provides full facilities for physically challenged students, as outlined.**

**Else, in proportion to facilities.**

- Basis: Verifiable Responses to Questions.
- ***Primary Data: To be provided in a prescribed format.***

## **5. Perception (PR) – 100 marks**

- **Ranking weight: 0.1**
- **Overall Assessment Metric: PR = 100**
- **Component metrics are explained in the following pages:**
  - A. Peer Perception: Employers & Academic Peers: (PR)**

**A. Peer Perception: Employers & Academic Peer (PR) : 100 marks**

- This is to be done through a survey conducted over a large category of Employers, Professionals from Reputed Organizations and a large category of academics to ascertain their preference for graduates of different institutions.
- Comprehensive list will be prepared taking into account various sectors, regions, etc.
- Lists to be updated periodically.